

Highlights

{TODO: Change title to reflect intro changes} SolidSessionBench: Realistic Query Sequences for User-Oriented Decentralized Environments

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- Research highlights item 1
- Research highlights item 2
- Research highlights item 3

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ARTICLE INFO

Keywords:

Multi-perspective data
Link Traversal-based Query Processing
Client-side caching
Query sequence simulation

ABSTRACT

Emerging decentralized architectures give users complete control over their own data, scattering information across heterogeneous sources. Without a central authority to enforce consistency, users freely publish overlapping and potentially conflicting claims, generating a highly multi-perspective decentralized environment. **RT: The jump between these sentences feels a bit too large. You could for example add an additional sentence here to talk about why integrating such decentralized data is challenging, and that client-side query engines can solve this.** While client-side query engines can dynamically fetch and reconcile this information, optimizing execution remains challenging due to a lack of prior knowledge regarding data distribution. By exploiting access patterns across repeated user query sequences, engines can overcome this knowledge gap. However, accurately evaluating these personalized optimizations is difficult. Public query logs for decentralized environments are scarce, and existing benchmarks rely on static, template-based structures that fail to capture realistic behavioral variability. To fill this gap, we introduce an evidence-based simulation of client query usage patterns as a reusable benchmark enabling realistic and reproducible evaluation of client-side optimization techniques. We demonstrate its utility by evaluating novel client-side caching strategies for link traversal query processing. By comparing our sequence-based approach against traditional, non-sequence-based workloads, we highlight how realistic sequence simulation heavily influences performance evaluations. Ultimately, our benchmark reveals important performance dynamics, such as the impact of session switching and query refinement, that traditional workloads do not. We conclude that modeling realistic user interactions provides an essential foundation for accurately evaluating client-side optimization techniques in decentralized environments.

Multi-perspective data Link Traversal-based Query Processing \end{keyword} which contain the abstract and keywords respectively.

Each keyword shall be separated by a \sep command.

1. Introduction

Emerging web applications increasingly rely on decentralized architectures to give users control over their personal data. Structured decentralized ecosystems, such as Solid [50] and Mastodon [46], encourage users to keep their data in small, independent stores following shared data models. This paradigm shift is increasingly vital for automated systems, such as LLM-based AI agents, which query these decentralized structures to extract facts for task performance or question answering [63, 33, 10]. While this approach strengthens user autonomy, the absence of a central data authority naturally creates a highly multi-perspective environment. Data spreads across thousands of heterogeneous sources that frequently contain overlapping, diverse, or potentially conflicting claims. Query engines must therefore dynamically fetch and integrate this information at runtime [55], not only to discover conflicting statements and multiple perspectives, but to reconcile these claims to provide a coherent answer to the user or agent.

Traditional federated querying approaches [53, 51, 31] support the integration of decentralized data sources, but they typically assume a small (10-100) set of well-described and consistently available endpoints [17]. These assumptions fail once data is spread across thousands of small units that enforce local access-control policies. In addition, enforcing fine-grained access control policies [20] on large sources with expressive interfaces introduces notable overhead [42], making standard approaches unsuitable for highly decentralized environments.

Newer decentralized environments, such as Solid, often expose data as lightweight HTTP resources instead of highly expressive query interfaces. Many of these resources are not known in advance and can only be discovered during query execution. While their simplicity lowers the cost of enforcing fine-grained controls, this setting requires a federation model that can operate over a very large number of small sources, many of which may be encountered at runtime, without relying on prior aggregation. Traditional federated approaches are insufficient in this setting due to the assumption of a small number of large endpoints. Additionally, these approaches often assume prior knowledge of the sources, which is not feasible in decentralized environments where discovery is a key requirement. Executing SPARQL queries using Link Traversal-based Query Processing (LTQP) [29] fits these needs. It discovers documents incrementally by

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following URIs encountered during execution, starting from an initial set of seed documents provided by either the user or the query. Its link-driven, asynchronous traversal supports fine-grained permissions and allows systems to discover without knowing the location of all decentralized data sources beforehand.

However, the fact that LTQP discovers documents incrementally creates significant challenges for efficient execution. Using the traditional optimize-then-execute query processing paradigm is difficult, as the query itself determines which data will be accessed. As a result, the optimizer has no prior knowledge of the relevant data until the query's traversal has already begun. Some approaches optimize query execution without prior knowledge by using zero-knowledge query planning [27] or adaptive query processing [25]. These yield modest but inconsistent performance gains, depending on the decentralized environment and the heuristics used [58]. However, other aspects of zero-knowledge LTQP optimization, such as link prioritization [30], are unlikely to improve query result arrival times in emerging decentralized environments [19].

Consequently, other works explore using precomputed server-side information as prior knowledge or relevance indexes to efficiently execute LTQP queries [59, 60, 24]. While these approaches offer greater performance benefits, they require the server to maintain these indexes. This raises the cost of serving resources and complicates privacy and access control.

To avoid server-side overhead, engines can derive prior knowledge directly on the client. In decentralized settings, LTQP operates inherently client-side, with a single engine instance serving one or more applications, often specific to a particular user. As these users and thus the used applications issue sequences of correlated queries, the engine identifies frequently accessed data and its characteristics. This allows engines to implement *personalized* query optimization algorithms that leverage local access patterns to build necessary prior knowledge.

{TODO: Adjust figure to reflect fact that AI is no longer focus} RT: In the figure, make sure to align the two tints of red. **RT:** I'd make it explicit somewhere that the different arrows indicate the traversal paths of different LTQP executions (each for one specific query).

Benchmarking these approaches requires realistic query sequences. Traditional benchmarks (e.g., WatDiv [1], BSBM [11], LargeRDFBench [49]) execute isolated or repeated query templates, which fails to capture client-side dynamics accurately. Furthermore, while centralized SPARQL optimization relies on real query logs [9, 44, 12, 54], such logs do not yet exist for decentralized LTQP environments. Consequently, we need a standardized benchmark that realistically simulates user query sequences to prevent fragmented evaluation practices.

In this paper, we address this gap by introducing SolidSessionBench, a benchmark extending SolidBench [58] to generate realistic query sequences. These sequences capture usage patterns identified in social network literature, a

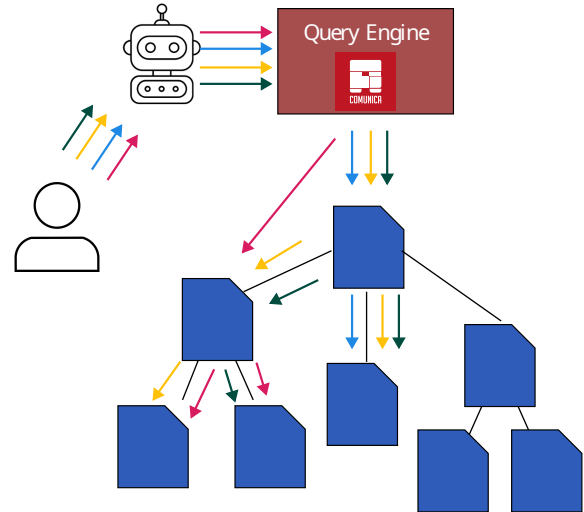


Figure 1: When users interact with agents, their queries intentions and thus relevant data to the query can overlap, leading to opportunities for personalized optimization.

domain naturally rich in multi-perspective data, alongside patterns observed in real-world SPARQL logs.

To evaluate the realism of our generated query sequences without access to real-world query logs, we investigate **{TODO: insert}** and evaluate **{TODO: insert}** baseline client-side caching strategies in the Comunica LTQP engine [56, 58]. We use these simple algorithms to establish a performance baseline and demonstrate the necessity of sequence-based evaluation. Comparing this baseline against existing literature shows that current evaluation techniques are insufficient for client-side caching in LTQP.

By enabling researchers to evaluate personalized query optimization against actual user behaviors, SolidSessionBench lays the foundation for performant client-side querying. Ultimately, efficient LTQP execution over decentralized sources facilitates the rapid identification of conflicting information and the application of consistency fixes across multi-dimensional knowledge graphs.

The remainder of this paper is structured as follows. Section 2 reviews prior benchmarking approaches and real-world query usage patterns. Section 3 outlines our simulation methodology and defines the resulting hyperparameters. Section 4 introduces the technical mechanics of client-side caching strategies in LTQP. Section 5 evaluates the benchmark using proxy measures of realism. Finally, Section 6 concludes the paper and presents a plan for benchmark sustainability.

{TODO: Maybe insert section summarizing contributions.}RT: Yes, that would be good!

2. Literature Review

To ground our benchmark design in existing research and ensure a realistic sequence generator, this section reviews

prior benchmarking approaches for client-side query optimization and real-world query usage patterns.

2.1. Existing Benchmarking Approaches

For the evaluation of client-side benchmarking techniques that rely on previous queries, we primarily investigate caching literature, as this is a well-studied problem relevant to our work.

In the literature, we find three distinct benchmarking approaches: simulated micro benchmarks, simulated end-to-end query benchmarks, and real-world query logs.

Simulated micro benchmarks attempt to isolate performance gains of the benchmarked approaches by fixing certain aspects of query execution. This can range from fixing the cached data size [26] to controlling the percentage of query-relevant data present in the cache [40]. Fixing aspects of query execution makes it easier to isolate and understand the effect of specific caching mechanisms. However, it also gives an incomplete picture, because we don't know how those fixed conditions relate to real workloads or how the methods perform when those assumptions don't hold.

Simulated end-to-end query benchmarks use generated query sequences to evaluate performance. They aim to measure cache hit rate and overall execution behavior under the assumption that the simulated sequences reflect realistic usage. In practice, the simulation is usually built by extending an existing benchmark so that it generates sequences instead of isolated queries. The sequence order is often determined by simple rules or by random selection.

A common approach is to use queries from existing benchmarks such as the Star Schema Benchmark [41] or the Berlin SPARQL Benchmark [11]. These queries are then randomly divided into sequences [26, 40, 22], with each sequence executed using a shared cache. This approach is straightforward but not realistic. Prior work shows that real workloads contain user behavior patterns and temporal locality, and these patterns can be exploited by caching methods. Random sequences fail to capture these properties.

More realistic is to adjust the query generators of existing benchmarks to include patterns in the query sequences. Some works [64, 36] use a Pareto distribution to decide what entity is used to instantiate a query template. This follows from the observation that many human activities follow a Pareto distribution [6], with many queries relating to a small portion of the entities.

A third approach uses simulated query sequences that reflect the access patterns of specific applications. One example is a workload that models data scientists exploring different aspects of the same entity—such as an author in a scholarly network—through consecutive, thematically related queries [15]. Such simulations often include multiple query sessions, with new sessions beginning according to a probability parameter p . Another example is a simulated application scenario focused on retrieving information about vacation destinations, which is used to assess caching effectiveness [36].

Real-world query logs Whenever available, query logs from actual applications form a highly relevant benchmarking approach. However, obtaining these query logs can be challenging. For SPARQL query optimization, query logs for a variety of endpoints are available, such as DBpedia [48, 54, 9] and Wikidata [54]. These logs are extensively used in benchmarking SPARQL-based approaches [65, 43, 34].

The evaluation of personalized LTQP optimization would ideally use real-world query logs, but such logs do not exist. To avoid the fragmented evaluation practices of current simulated approaches, the next best option is a unified simulated benchmark. A shared benchmark would provide a common basis for comparison, improve reproducibility, and prevent each paper from recreating its own evaluation setup.

2.2. Real-world Query Usage Patterns

To inform the benchmark design on what patterns should be simulated, we will outline three relevant client query usage patterns observed in the literature. The relevance is determined for our social network application use case.

User Interest-Based Query Behavior Work on social networks shows users form communities based on shared interests [38, 16, 7, 32]. These networks exhibit power-law degree distributions, small-world structures, and scale-free properties. This topology creates a dense core of high-degree nodes that connect many smaller, strongly clustered groups [38]. Simulation studies confirm that this interest-driven group formation is stable and reliably modeled [2, 23].

Furthermore, empirical research formally establishes that shared interests directly drive a higher degree of explicit and implicit interactions between users [39]. These findings justify our simulation of user-relatedness and interest-driven behavior when generating query sequences.

Logical Query Sessions Analysis of web browsing behavior shows that activity is best grouped into *logical sessions*: sequences of user interactions focused on specific goals, rather than purely time-bound intervals [37, 47, 52]. These task-oriented sessions frequently involve non-linear navigation, including backtracking. New sessions begin in approximately 15% of clicks, following a log-normal distribution with average size and depth means of 6.1 and 3, respectively.

These patterns also appear in application navigation. In a decentralized social media application, for example, a user viewing a feed might click a creator's username, view comments, or like a post. Because the parameters of each new query derive directly from the preceding query's results, these queries chain together into structures mirroring logical web sessions. Additionally, direct searches for specific posts or topics reflect standard session-switching behavior.

Consequently, interactive applications generate sequences of interdependent requests [62]. This dynamic informs the design of Facebook's TAO data store, which is explicitly optimized for workloads driven by the user's step-by-step traversal of the social graph within the interface [14].

Query Refinement Analysis of SPARQL queries issued to the DBpedia endpoint reveals that users frequently submit related queries in streaks [13, 66]. These streaks consist of query sequences that share an underlying intention but undergo iterative refinement to achieve the user’s objective. Previous studies initially introduced the concept of SPARQL query sessions specifically to analyze robotic queries [66, 13].

In this benchmark, we focus on organic query sequences [66], which users generate through interactions with software or browsers [35]. Because organic interactions are driven by application design rather than underlying data organization, we assume these user behaviors will also emerge in applications navigating structured decentralized environments. Furthermore, we deprioritize robotic queries because their sequences can already be simulated by traditional benchmark approaches [58, 11, 1].

For both organic and robotic queries, analysis of total usage and reformulations (removals and additions) of operators over all query logs shows that filter, union, and optional operators account for the vast majority of reformulated queries.

Recent analyses of Wikidata query logs further demonstrate that query development is an iterative process of design, debugging, and maintenance [5]. Based on this empirical data, exploratory querying encompasses six core behaviors [5]: **result refinement** to reduce records via filters or selective triple patterns, **result generalization** to retrieve more entities by broadening parameters, such as adding superclasses or increasing limits, **result extension** to fetch additional attributes by appending new triple patterns, **bug fixing** to correct structural errors like invalid URIs, **query refinement** to adjust complex language features, and **shifting query intent** to pivot topics by altering core predicates.

These findings confirm that real-world querying relies on dynamic, step-by-step modifications rather than the static execution of independent templates. Consequently, existing evaluation frameworks fail to capture how users actively interact with data systems. Evaluating system performance for these exploratory query sequences remains a critical research challenge, explicitly requiring new benchmarks to validate progress [5]. Therefore, modeling these empirical sequence patterns is essential to accurately assess client-side query optimization techniques.

3. Query Sequence Benchmark

In this section, we detail our simulation methodology and the resulting hyperparameters. We base our sequence generator on SolidBench [58], which simulates realistic decentralized social media applications within the Solid ecosystem. SolidBench utilizes the Social Network Benchmark (SNB) dataset [18, 3] and applies the Linked Data Benchmark Council (LDBC) choke point methodology [4]. **RT: We should explain this methodology (somewhere).** Its workload combines standard SNB *interactive* queries with custom *discover* templates, which explicitly

target Link Traversal Query Processing (LTQP) choke points in decentralized environments.

To support sequence-based evaluation, we extend three core components of the SolidBench ecosystem: **RT: We should make sure to properly explain SolidBench and its architecture (we can do that now, with the space limitations of a journal).**

- **Query Generator:** We modify the generator to produce correlated query sequences that reflect empirical real-world usage patterns.
- **Dataset Generator:** We enhance the data generation process to model interest-based similarities between simulated users.
- **Benchmark Runner:** We integrate sequence execution logic into the existing runner to facilitate straightforward, reproducible sequence-based experiments and allow easy generation of default configuration files.

3.1. Realism of Query Sequences

To ensure SolidSessionBench accurately reflects user behavior in decentralized environments, we base our simulation methodology on the LDBC methodology used for data generation. The sequence generator synthesizes empirical usage patterns established in Section 2 to form a grounded simulation methodology that reflects real-world patterns.

Because direct, real-world query logs for highly decentralized LTQP environments do not currently exist, we maintain and evaluate realism through a three-pillared approach:

- **Topological Realism (User Interests):** We utilize the established LDBC social network topology to model explicit and implicit user relatedness, ensuring data access patterns mirror natural community formations rather than uniform distributions. Due to the proven realism of the LDBC data generator [45], our simulations inherit this empirically proven structure.
- **Navigational Realism (Logical Sessions):** We restrict query sequences to directed dependency graphs. This simulates realistic “click-through” application navigation, accounting for empirically observed session lengths and log-normal transition probabilities.
- **Exploratory Realism (Query Refinement):** We replace static templates with dynamic query transformations. Mapping real-world SPARQL refinements (e.g., result generalization, intent shifting) directly to execution choke points ensures engines face realistic structural variability.

We evaluate the success of this simulation by measuring proxy variables such as cache hit rates, execution times, and relatedness of relevant data accessed during queries. We then compare these outcomes against theoretical baselines and random sequence generation. This comparative evaluation demonstrates that our query sequence simulation captures

essential performance dynamics that existing benchmarks miss.

3.2. Simulation Methodology

Our methodology models three interdependent behaviors: task-oriented logical sessions, user interests for data locality, and refinement patterns for iterative query development.

3.2.1. Logical Query Sessions

To model logical sessions, we categorize queries into four primary topics: person-specific messages, forum information, person attributes, and message attributes. Because queries in decentralized environments often depend on preceding results, we map the output types of each template to valid subsequent templates. This creates a directed graph of query progression that governs the flow of a logical session.

Within a session, a template's *instantiation values* are typically derived from the previous query's results. To realistically simulate this "click-through" behavior, we execute centralized equivalents of the LDBC SNB queries using QLever [8]. We then randomly sample the returned result set to instantiate the subsequent query template. If a query yields no results, we fall back to our default, interest-based instantiation methodology.

As an example, consider the following data distributed across a decentralized environment:

Listing 1: (a) Alice's profile document

(<https://pod.example/alice/profile.ttl>).

```
1 <https://pod.example/alice/profile.ttl#me> a :
    Person ;
2 :knows <https://pod.example/bob/profile.ttl#me>
    ;
3 :hasInbox <https://pod.example/alice/inbox> .
```

Listing 1: (b) Bob's profile document

(<https://pod.example/bob/profile.ttl>).

```
1 <https://pod.example/bob/profile.ttl#me> a :Person
    ;
2 :posts <https://pod.example/bob/posts/1> .
```

Listing 1: (c) Bob's post

(<https://pod.example/bob/posts/1>).

```
1 <https://pod.example/bob/posts/1> a :Post ;
2 :content "Hello Solid world!" .
```

Listing 1: (d) Alice's interests document

(<https://pod.example/alice/interests.ttl>).

```
1 <https://pod.example/alice/profile.ttl#me> :
    interestedIn "Wildlife" .
```

Listing 1: (e) Wildlife forum document

(<https://pod.example/forum/wildlife.ttl>).

```
1 <https://pod.example/forum/wildlife.ttl> a :Forum ;
2 :hasPost <https://pod.example/ruben/posts/9> ;
```

```
:hasComment <https://pod.example/bob/comment/1>
    ;
4 :hasTopic "Wildlife" .
5 <https://pod.example/bob/comment/1> :responseTo <
    https://pod.example/ruben/posts/9> .
```

The generator randomly selects a user and simulates a realistic sequence of queries that reflects real-world user behavior. Given this dataset, and two query templates: one to fetch a person's posts and another to retrieve the content of a specific post, the generator can create a query sequence, which can contain multiple logical sessions. For user "alice", we can start the sequence (and first logical session) by executing the first query template (Q1) to discover Bob's posts, as shown in Listing 2.

Listing 2: Initial query (Q1) fetching Bob's posts.

```
SELECT ?post WHERE {
  <https://pod.example/bob/profile.ttl#me> :posts ?
    post .
}
```

The client-side query engine executes this query by traversing to Bob's profile document, returning the result binding: '?post = <https://pod.example/bob/posts/1>'.

To simulate a user clicking on this discovered post, the sequence generator instantiates the next query template dynamically. It extracts the value bound to "?post" from the results of Q1 and injects it as the subject of Q2, as shown in Listing 3.

Listing 3: Dynamic sequence query (Q2) instantiating a variable from Q1.

```
SELECT ?content WHERE {
  <https://pod.example/bob/posts/1> :content ?
    content .
}
```

This example demonstrates that the instantiation of Q2 strictly depends on the execution results of Q1. Because Q2 requires an instantiated post URI from Q1, these queries are directionally dependent; reversing their order leaves the generator with no method to instantiate the initial query parameters. Consequently, structured logical query sequences naturally emerge from these interdependencies, accurately reflecting realistic exploratory user behavior.

Finally, we simulate logical session switching. Drawing from web browsing behavior, we assign each sequence a session-switching probability sampled from a log-normal distribution with a mean of 15% [37, 52]. During sequence generation, a session switch triggers a random choice with equal probability: the engine either starts a new session (using interest-based instantiation) or resumes a previous one (deriving instantiation values from that session's last executed query). To prevent repetitive queries and maintain a uniform distribution of template instantiations, we explicitly prohibit backtracking within our model. Additionally, we maintain a balanced template distribution by dynamically scaling down each template's selection probability proportionally to its overall frequency.

Continuing our running example, a session switch occurs after executing Q2. Instead of continuing sequence generation from the content of Bob's post, the generator terminates the current path and selects a new template to instantiate: a query for posts within a forum.

As shown in Listing 4, the generator selects a wildlife forum as the initial instantiation value. This query requires an entirely separate data path to execute, breaking the dependency chain of the previous queries and representing a clear change in user intent. We are still generating the same *query sequence*, but the session switch introduces a new *logical session* within that sequence.

Listing 4: Independent query (Q3) initiating a session switch to explore a wildlife forum.

```
1 SELECT ?post WHERE {
2   <https://pod.example/forum/wildlife.ttl> :hasPost
3     ?post .
4 }
```

3.2.2. User Interest Modeling

Individuals interact more frequently with users sharing similar interests [39]. To simulate this behavior, we use the underlying LDBC dataset utilized by SolidBench [18], which inherently models realistic user preferences. The LDBC generator, Datagen, produces person profiles containing specific interests, educational backgrounds, and workplaces. It links users based on these demographic and interest overlaps, yielding a non-random network topology with a realistic degree distribution.

We use this interest-driven topology to generate template instantiation values for the first query of each logical session. Specifically, we construct a binary feature vector v_k for each user k :

$$v_k = (x_{k,1}, \dots, x_{k,n}, y_{k,1}, \dots, y_{k,m}), \quad (1)$$

where $x_{k,j} = 1$ if user k holds stated interest j , and $y_{k,m} = 1$ if user k completed study m .

For each sequence, we randomly select a base person k and compute the cosine similarity between v_k and the feature vectors of all other users. Let N be the total number of users. To maintain efficiency at large scale factors, we restrict our candidate pool to $C_{k,M}$, defined as the set of the top $M < N$ users ranked by similarity to k . Template instantiation values are then selected probabilistically using a softmax function with temperature T :

$$P(i | k) = \frac{e^{s_{k,i}/T}}{\sum_{u \in C_{k,M}} e^{s_{k,u}/T}}, \quad (2)$$

where $s_{k,i}$ is the similarity score between base person k and candidate $i \in C_{k,M}$. For templates requiring non-person values (e.g., posts or activities), selection probabilities are derived from the similarity scores of the content's creators.

Continuing our running example, consider an expanded decentralized environment containing a second forum dedicated to pottery, as shown in Listing 5 as well as all documents in Listing 1.

Listing 5: The additional pottery forum document (<https://pod.example/forum/pottery.ttl>).

```
1 <https://pod.example/forum/pottery.ttl> a :Forum ;
2   :hasTopic "Pottery" .
```

To determine what instantiation value to use for the forum query for user Alice, we construct binary feature vectors over a small global feature space: (Wildlife, Pottery).

Alice's profile states an explicit interest in 'Wildlife', yielding vector $v_{\text{Alice}} = (1, 0)$. While the forums in the data have topic statements, thus the vectors for the forums are $v_{\text{Wildlife}} = (1, 0)$ and $v_{\text{Pottery}} = (0, 1)$.

This gives us a cosine similarity of $s_{\text{Alice,Wildlife}} = 1$ and $s_{\text{Alice,Pottery}} = 0$. Then using softmax with a temperature of $T = 0.5$, we compute the selection probabilities for the two forums:

$$P(\text{Wildlife} | \text{Alice}) = \frac{e^{1/0.5}}{e^{1/0.5} + e^{0/0.5}} \approx 0.88,$$

$$P(\text{Pottery} | \text{Alice}) = \frac{e^{0/0.5}}{e^{1/0.5} + e^{0/0.5}} \approx 0.12.$$

Consequently, when initiating or switching a session, the generator is more likely to select the 'Wildlife' forum for Alice, reflecting her interests.

3.2.3. Refinement Patterns

The benchmark simulates query refinement by applying composable transformations to an instantiated query template. Following our literature review, we focus on transformations (and their reciprocal counterparts) that add or modify BGPs, OPTIONAL clauses, UNION blocks, and query instantiation values.

To ensure these transformations meaningfully alter query execution rather than superficially appending triples, we apply a choke point-based design methodology [4]. The generator sequentially applies refinements to a base SolidSessionBench template. By mapping empirical exploratory usage patterns [5] to concrete planning (P) and traversal (T) choke points, we translate real-world user behaviors into rigorous execution bottlenecks for query engines.

The planning choke points, derived from empirical usage patterns [5] and relevant LDBC choke points [3], are defined as follows:

- P.1** Involves adding or removing a FILTER, requiring prior knowledge to adjust query plans and estimate selectivity.
- P.2** Addresses adding or removing UNION or OPTIONAL clauses to alter filter push-down capabilities and complicate query planning.
- P.3** Covers adding or removing selective triple patterns, forcing the engine to identify changes and re-optimize the plan.
- P.4** Focuses on adding or removing non-selective triple patterns.

Beyond planning, these refinements introduce traversal choke points that expand upon those established in SolidBench [58, 57]:

- T.1** Expands or contracts the reachable sub-web through predicate additions.
- T.2** Shifts the sub-web by substituting the traversal starting point.
- T.3** Changes the number of hops required to obtain relevant data, corresponding to SolidBench choke points L.1–L.6 [57].
- T.4** Alters the usability of structural indexes, such as type indexes, corresponding to SolidBench choke point L.8 [57].

These choke points directly map to the exploratory query behaviors identified in Section 2.2:

- **Result refinement** corresponds to P.1 and P.3, as it narrows results via FILTER constraints and selective triple patterns.
- **Result generalization** corresponds to P.3 and P.4, expanding existing BGP patterns with selective or non-selective triple patterns.
- **Result extension** maps to P.2, P.3, and P.4 by appending triple patterns to BGP patterns or introducing OPTIONAL and UNION blocks that alter filter push-down capabilities.
- **Shifting query intent** corresponds to T.1 and T.2, substituting instantiation values to initiate traversal from different seed documents, thereby shifting or expanding the reachable sub-web.
- **Query refinement** maps to P.2, modifying the structural complexity of the query by adding or removing OPTIONAL and UNION blocks.

Finally, the generator supports variable instantiation within refinements, allowing users to bind pattern variables to specific values randomly selected from configuration candidates. Users can define custom refinement patterns via JSON files to augment the default configurations.

An example of a slimmed-down default configuration is provided in Listing 6. This entry specifies an OPTIONAL refinement pattern that appends a triple pattern to retrieve the messages liked by a person.

The location parameter is set to 0, indicating that the triple pattern is added to the first existing OPTIONAL block, or to a new block if none is present. The default benchmark suite also includes a reciprocal configuration where operation is set to “removal”, which instead deletes the triple pattern from the first OPTIONAL block.

Listing 6: Example configuration for an OPTIONAL refinement pattern.

```
1 {
2   "type": "OPTIONAL",
```

Table 1

LogT-normal distribution parameters (μ, σ) control the queries per sequence, queries per logical session, and the probability of switching logical sessions. Refinement parameters define the likelihood (patternProbability) and bounds (min/maxRefinementLength) of generating a refinement sequence for a given query. Instantiation relies on a softmax temperature and a maxLogits cutoff for similarity-based entity selection.

Category	Parameter	Value
Distributions (μ, σ)	Sequence Length	3.0, 0.2
	Session Length	1.7, 0.5
	Transition Probability	-2.0, 0.25
Refinements	patternProbability	0.1
	min/maxRefinementLength	2, [3–5]
Instantiation	temperature	0.1
	maxLogits	5–10

```
3   "id": 0,
4   "operation": "addition",
5   "description": "Add triple for likes (P.1, T.2)",
6   "location": 0,
7   "target": [
8     {
9       "subject": {
10        "value": "person",
11        "termType": "variable"
12      },
13      "predicate": {
14        "value": "../vocabulary/likes",
15        "termType": "namedNode"
16      },
17      "object": {
18        "value": "likedMessage",
19        "termType": "variable"
20      }
21    }
22  ]
23 }
```

All defaults, along with complete documentation, are available on GitHub.¹

3.3. Hyperparameters

The sequence generation process introduces several new hyperparameters, which are detailed in Table 1 together with their default values. To guarantee reproducibility despite extensive random sampling, a single seeded pseudo-random number generator (PRNG) is used throughout the entire pipeline.

4. Caching in Link Traversal-based Query Processing

Unlike traditional query engines that evaluate queries against a centralized, pre-built index, LTQP evaluates queries

¹<https://github.com/SolidBench/SolidBench.js/tree/master/templates/refinements>

directly over the live Web [57, 28] using the “follow-your-nose” principle.

Starting from a seed URI (often extracted from the query), the engine fetches data via a *link queue* (governed by *reachability criteria* [29]) and deposits the retrieved triples into an *active local store*. This local store is an in-memory dataset over which the query executes, typically utilizing dictionary encoding [56].

Previous work [26] established the fundamental concept of caching within this paradigm, demonstrating that a local repository of recently accessed Web documents mitigates the high network latency inherent to LTQP.

For our experiments, we use the Comunica LTQP engine [56, 58]. As a JavaScript-based engine, it relies on the event loop to asynchronously interleave HTTP requests during traversal and local query execution.

We use caching-based algorithms to validate SolidSessionBench query sequences and compare them against alternative benchmarking methods in the literature. Implementing these strategies establishes strong baselines for evaluating caching in traversal engines. This section introduces four cache implementations, each storing dereferenced triples differently. We also evaluate two cardinality estimation approaches that use cached triples to improve query planning and execution. Because these estimates require an indexed cache, we omit them for the unindexed baseline.

4.1. Cache Implementations

Unindexed LRU Cache This baseline caches raw triple arrays alongside source metadata. On a cache hit, the engine retrieves and indexes these triples into the active local store.

Indexed LRU Cache This approach pre-indexes triples and stores them in an in-memory LRU cache. On a hit, the engine extracts only the triples matching the query patterns. Combined with dictionary encoding, this pre-filtering reduces local store memory and indexing overhead, potentially offsetting the higher initial ingestion costs.

Memory-based Quad Store Cache This approach uses a quad store to unify cache indexes, saving memory and improving search efficiency. The quad graph stores the source of the triples, allowing cache hits to retrieve data using an ‘?subject ?predicate ?object’ query where the graph term matches the cached document’s URL.

Disk-based Quad Store Cache This approach stores the quad store on disk, providing the same advantages while supporting datasets larger than main memory. We add a small in-memory cache for a ‘hot’ subset of frequently accessed triples. To make this memory cache scan-resistant, documents are only added if accessed multiple times. This prevents long-running queries from evicting the entire cache.

4.2. Cardinality Estimation Approaches

Link traversal lacks prior knowledge regarding data distribution and availability, complicating query planning. To

address this, we hypothesize that cached data provides sufficiently accurate cardinalities for query planning, and we evaluate two estimation approaches based on this cache. We integrate these estimates into the default Comunica cost-based optimizer [56] to generate more accurate query plans, replacing the zero-knowledge heuristics [27] traditionally used in link traversal.

Reachability-agnostic Cache-based Cardinality Estimation This approach estimates cardinalities across the entire cache. For each cache entry, it performs a query to count the number of matching triples and sums these counts to produce a final estimate.

Reachability-aware Cache-based Cardinality Estimation Here, we store the outgoing links of each cached document. For each query, we traverse the cache to find all documents reachable under the active reachability criterion, to produce more accurate estimates by ignoring irrelevant data. If the seed document is missing from the cache, we fall back to the Reachability-agnostic Cache-based Cardinality Estimation.

4.3. Experimental Configurations

For our experiments we evaluate three cache capacities: small (350k triples, ~10% of the dataset), medium (700k, ~20%), and large (1.4M, ~40%). While the disk-based cache stores 1.4M triples on disk, and a small in-memory cache of 25k, 50k, and 100k triples for the small, medium, and large configurations respectively.

Our implementations adhere to the RFC-9111 specification [21], meaning the cache correctly interprets standard HTTP headers (e.g., Cache-Control) just like a web browser would. However, because the benchmark dataset is static, cache entries require no revalidation and never become stale. Consequently, observed hit rates are likely higher than in a live environment.

Ultimately, comparing these strategies establishes 10 baselines: three algorithms per indexed cache (LRU, In-memory Quad Store and Disk-based Quad Store) and one algorithm for the unindexed cache, for caching performance in realistic user-centric environments, upon which future research into advanced caching approaches can build.

5. Evaluation

{TODO: Improve naming algorithms to new names we selected} {TODO: Here outline also why we evaluate the benchmark this way, why this is a proxy for a useful benchmark and then repeat again: no empirical data.}

We evaluate our benchmark using a scale 0.1 SolidBench dataset (158,233 RDF files, 1,531 vaults, 3,556,159 triples). Using the default hyperparameters in Table 1, we generate 8 sequences totaling 192 queries. These queries are from the interactive workload, are split into discover and short queries, with multiple templates belonging to either of these groups. Experiments run on an Ubuntu 20.04 machine (Intel Core i5-9400) with 16 GB RAM allocated to the engine

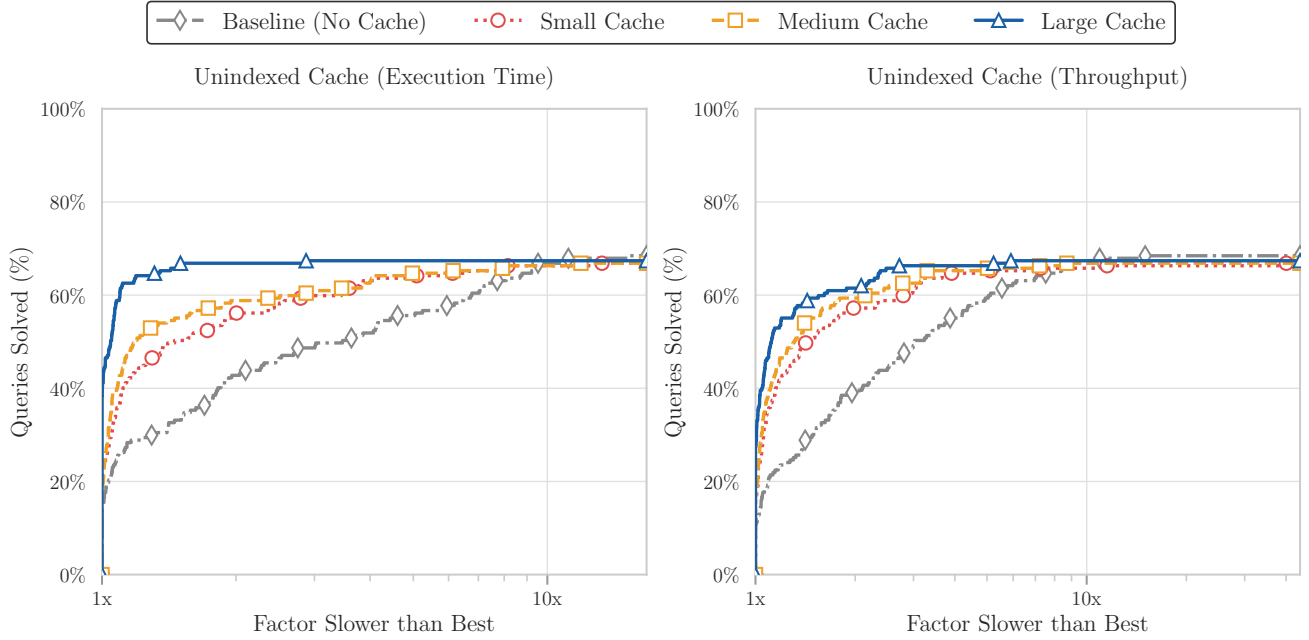


Figure 2: Performance profiles of the LRU-based unindexed triples cache. The y-axis shows the percentage of queries executing within a factor (x -axis) of the fastest algorithm. Curves closer to the top-left indicate better overall performance. The y-intercept ($x = 1$) shows how often an algorithm is the outright fastest, while the rightward trajectory demonstrates robustness (e.g., $x = 2, y = 80\%$ means 80% of queries execute within twice the fastest time). From the figure, we observe that all caching-based approaches outperform the no-cache baseline.

and a 180-second query timeout. We repeat each sequence 10 times to minimize variance and inject a uniform 70–90 ms latency to simulate a realistic network environment. The cache is scoped for a sequence, meaning no cache-state is shared between different executions or repetitions of sequences.

5.1. Holistic Evaluation of Benchmark Characteristics

First, we present the overall performance characteristics of the tested caching algorithms on the default configuration of our introduced benchmark.

Unindexed LRU Cache In Figure 2, we find that the unindexed LRU cache reliably improves performance across all cache sizes, with larger caches yielding better results. For total execution time of queries, the large cache is significantly better than the smaller cache sizes, while for throughput the difference is less pronounced.

Indexed LRU Cache As shown in Figure 3, pre-indexing triples in the LRU cache accelerates successful query execution but introduces higher initial overhead. When high cache eviction rates outweigh the efficiency gains of indexing, this ingestion cost slows down execution or causes timeouts. The figure illustrates this overhead, as the no-cache baseline ultimately solves a higher percentage of queries than the standard indexed cache as the line reaches a higher y -value.

However, integrating cardinality estimation overcomes this limitation by enabling the query engine to generate

more accurate query plans, increasing the percentage of solved queries and lowering execution times. Comparing the two estimation approaches, reachability-aware estimation outperforms reachability-agnostic estimation by ignoring irrelevant data. This effect is most pronounced in larger caches, where reachability-agnostic estimation yields only minor performance benefits, while reachability-aware estimation shows substantial improvements. This aligns with expectations: larger caches accumulate more irrelevant, out-of-context data, which the reachability-aware estimator successfully filters out.

Memory-based Quad store Cache As shown in Figure 4, the memory-based quad store cache exhibits high performance variance depending on the estimation strategy. Without estimation (*Indexed Store*), performance scales normally with capacity: the large cache achieves the best performance, while the medium and small caches sit lower but still reliably outperform the baseline. Similar to the LRU-based indexed cache, the baseline solves slightly more queries, indicating that the performance overhead can cause query failures. However, introducing online cardinality estimation (*Indexed Store (Estimate)*) completely inverts the performance scaling observed without estimation, with the small and medium caches outperforming the large cache.

The scaling nature of the performance penalty of cache size in estimation indicates that executing count queries scales unfavorably as the cached quad store size increases. Compared to the LRU-cache, which executes count triple pattern queries over small sources, the quad store-based

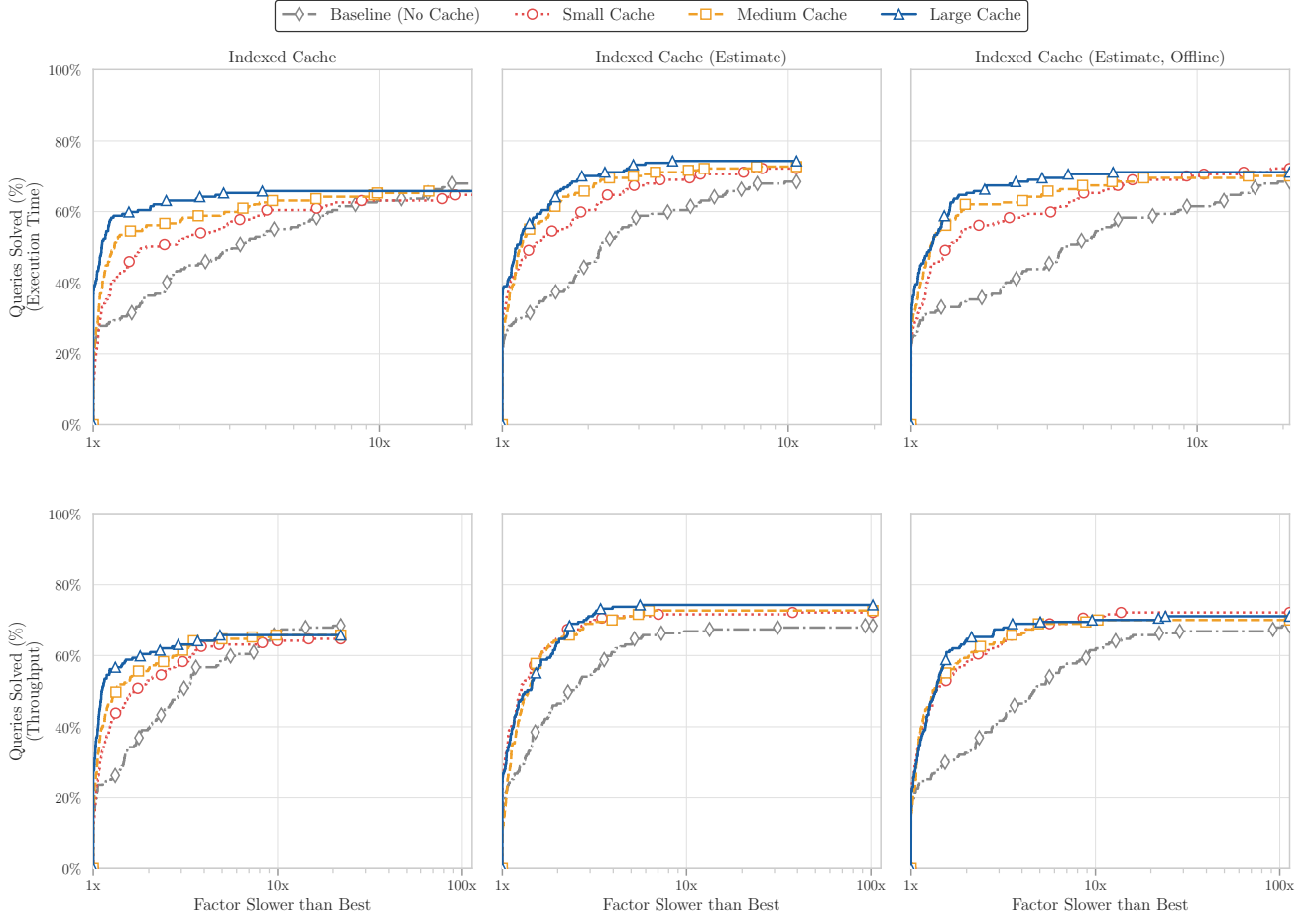


Figure 3: Performance profiles of the LRU-based indexed cache family of algorithms. The y-axis shows the percentage of queries executing within a factor (x-axis) of the fastest algorithm.

approach uses triple pattern queries over the entire quad store, but with the graph term filled in to represent the current document. For reachability-aware estimation the performance hit is not as big, especially in the total execution time case.

Disk-based Quad store Cache In Figure 5, we see that the disk-based quad store cache performs.... **{TODO: PLACEHOLDER FIGURE}**

Comparison Between Caching Algorithms **{TODO: Small section comparing the different caching algorithms and their performance characteristics. for example how unindex cache mainly improves total execution time, how quad store seems best when using small caches and LRU-cache scales better for bigger caches.}**

5.1.1. Cache Hitrates

{TODO: Optional section with hitrates and eviction percentages per cache size and algorithm.}

5.2. Logical Session & Interest Modeling Cache Performance

Present a table showing the effect of logical sessions on cache performance. Include a metric for the delta number of unique data sources accessed during session switches to quantify data locality. Discuss the implications of these session dynamics for cache sizing and eviction policies.

5.3. Refinement Patterns

{TODO: Write introduction to this section}

5.3.1. Effect of chokepoints on query sequences

{TODO: This has a lot of repetition yes?} We isolated the structural impact of individual query chokepoints through a controlled experiment comprising 80 single-query sequences. Each sequence begins with a base query and sequentially applies a chain of two to five refinement patterns from the benchmark's default refinement pattern set. We repeated each sequence execution 10 times across the three distinct cache capacities (Small, Medium, and Large), and measure the cache hit rate, the number of unique data sources accessed, and the total execution time. For this experiment we only use the LRU-based indexed cache, as we want to analyse the chokepoints and their effect on the cache in the

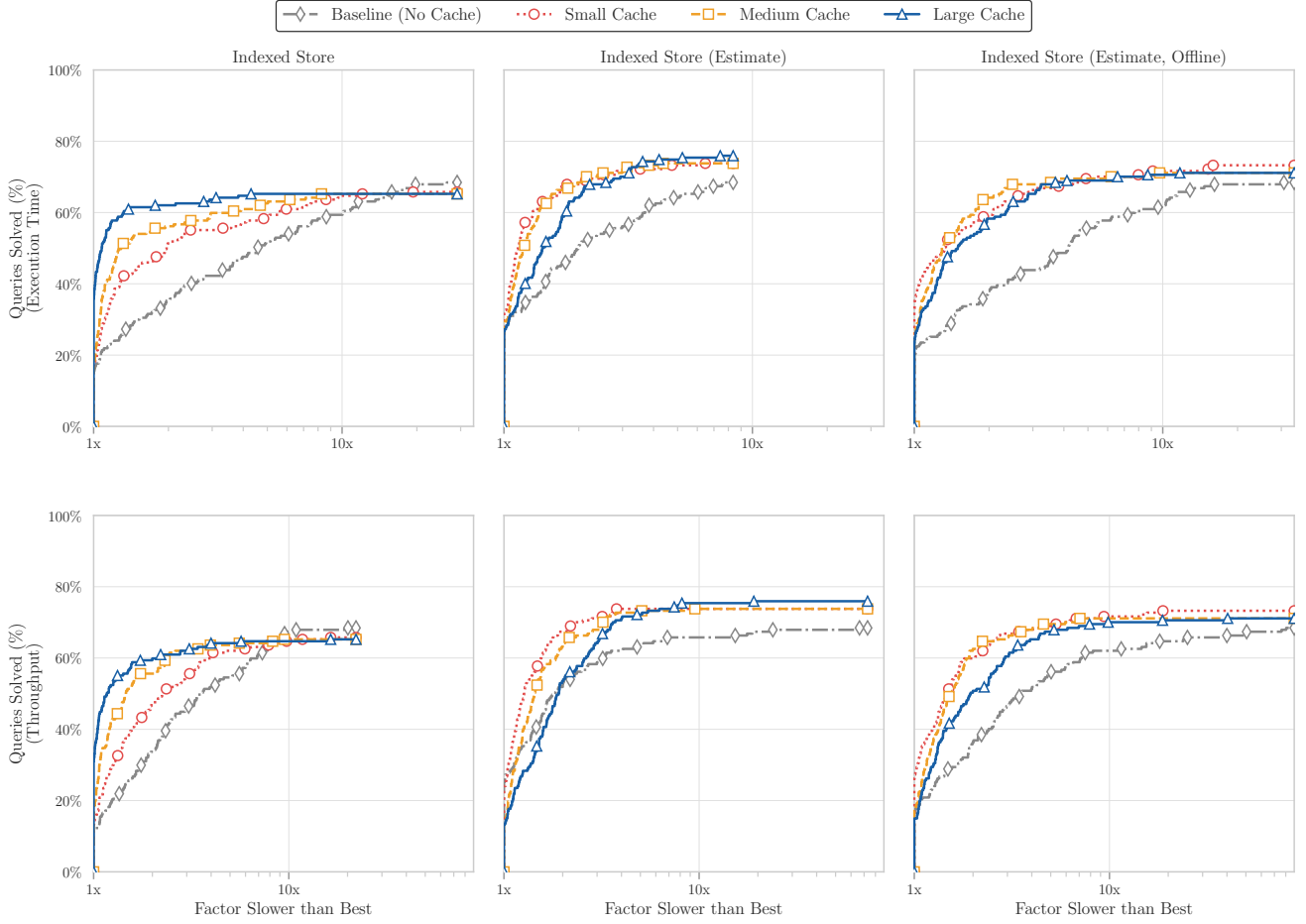


Figure 4: Performance profiles of the in-memory quad store cache family of algorithms. The y-axis shows the percentage of queries executing within a factor (x-axis) of the fastest algorithm.

simplest configuration. To quantify the specific performance impact of each chokepoint, we mapped the refinement patterns to their underlying chokepoints and evaluated them using a linear mixed-effects regression model. The model predicts the performance delta (ΔY) between the current and previous query step. We categorize chokepoint operations into two classes: additions (denoted by an ‘a’ suffix) and removals (denoted by an ‘r’ suffix). A removal operation only occurs when the sequence reverts a previously applied refinement pattern, thereby dropping the associated chokepoint from the query. The chokepoints added in a given refinement step are represented as a binary vector X_t , where each element indicates the presence (1) or absence (0) of a specific chokepoint. These vectors are then used as predictors in the regression model, in combination with the previous query’s performance metrics (Y_{t-1}) to account for the baseline state of the system. The full regression model is expressed as follows:

$$\Delta Y_{i,t} = \gamma Y_{i,t-1} + \mathbf{T}_{i,t}^\top \boldsymbol{\beta} + u_i + \epsilon_{i,t}, \quad (3)$$

where $\Delta Y_{i,t}$ is the performance delta for sequence i at refinement step t , $Y_{i,t-1}$ is the previous steps independent variable

value, $\mathbf{T}_{i,t}$ is a binary indicator vector of the applied chokepoints, $\boldsymbol{\beta}$ is the vector of fixed effects estimating the isolated impact of each pattern, u_i is a sequence-specific random intercept absorbing unobserved heterogeneity across base queries, and $\epsilon_{i,t}$ is the standard residual error.

Figure 6 shows the estimated effect of each chokepoint on performance. Full regression results are in Appendix ?? **[TODO: Verify appendix or supplemental material]**.

We observe two types of chokepoint behavior. First, modifying some patterns hurts performance in both directions. For example, both adding and removing τ_3 (increasing required traversal hops) reduces the hit rate and increases execution time. This reveals a strong cache invalidation penalty: changing the refined query breaks the cache, overwriting much of it and evicting useful data. From this, we can observe that caching requires a *scan-resistant* algorithm.

Second, patterns like τ_2 (new instantiation values) and P_4 (non-selective triple patterns) have reversible effects. Adding these chokepoints drops performance, but removing them improves the hit rate and speeds up execution. In these cases, the cache is able to maintain useful data across the refinement, and the chokepoint itself is the primary cause of

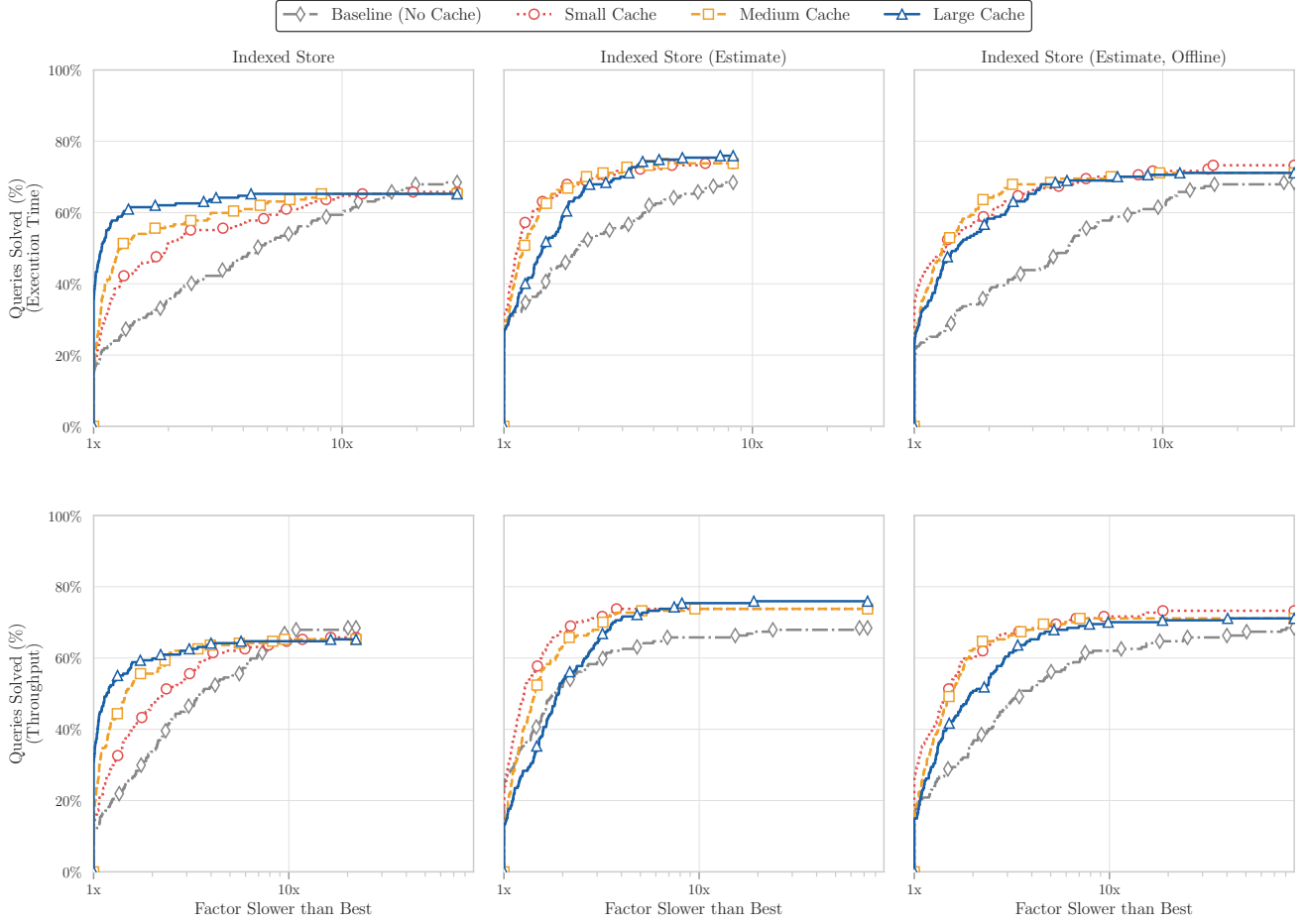


Figure 5: Performance profiles of the disk-based quad store cache family of algorithms. The y-axis shows the percentage of queries executing within a factor (x-axis) of the fastest algorithm. **{TODO: PLACEHOLDER FIGURE}**

performance degradation. Finally, across all patterns, chokepoints that increase the number of accessed data sources consistently lower the cache hit rate.

Implications for Refinement Pattern Design From our analysis, we find important weaknesses in our tested caching algorithms. We demonstrate that structural mutations like τ_3 trigger severe cache invalidation penalties, heavily degrading performance. Conversely, the reversible nature of patterns like τ_2 proves that caches can successfully maintain temporal locality even when the instantiation value, and thus the traversal starting point, changes. These findings highlight the importance of our refinement sequence generator and the mapping of refinement patterns to chokepoints. This mapping allows us to isolate the structural impact of each chokepoint and thus analyse the strengths and weaknesses of a given caching algorithm. Furthermore, because our sequence generation is dynamically configurable, it serves as a targeted diagnostic tool for future systems research. Developers can easily extend our default configurations to isolate specific chokepoints, allowing them to custom-build stress tests that target the unique operational challenges of their own caching architectures.

The resulting coefficients are given in Figure 6, where the x-axis represents the estimated effect size of each chokepoint on the performance delta, and the y-axis lists the chokepoints. For full regression results, we refer to the **{TODO: Where to put them?}**. From the figure we find that for some chokepoints, such as T.3 (increasing the number of hops required to obtain the data), both addition and removal significantly reduce cache hit rate and increase execution time. Indicating that the chokepoint causes the cache to evict previously cached data, which causes it to underperform when the chokepoint is removed. Another type of chokepoint, is where the addition of the chokepoint reduces the hitrate and increases execution time, while removing it increases hitrate while reducing execution time, such as T.2 (new instantiation value) and P.4 (adding non-selective triple patterns). Counterintuitively, for chokepoints such as ‘P.3.r’, ‘P.2.r’, and ‘P.1.a’, the large cache has a smaller positive effect on hitrate compared to the smaller caches. As cache hitrate is bound by 1.0, the large cache will have less room for improvement compared to smaller caches, which explains the smaller observed effect. This is supported by the average hitrates over all queries, which are 0.51, 0.61, and 0.66 for the small, medium, and large caches respectively and the

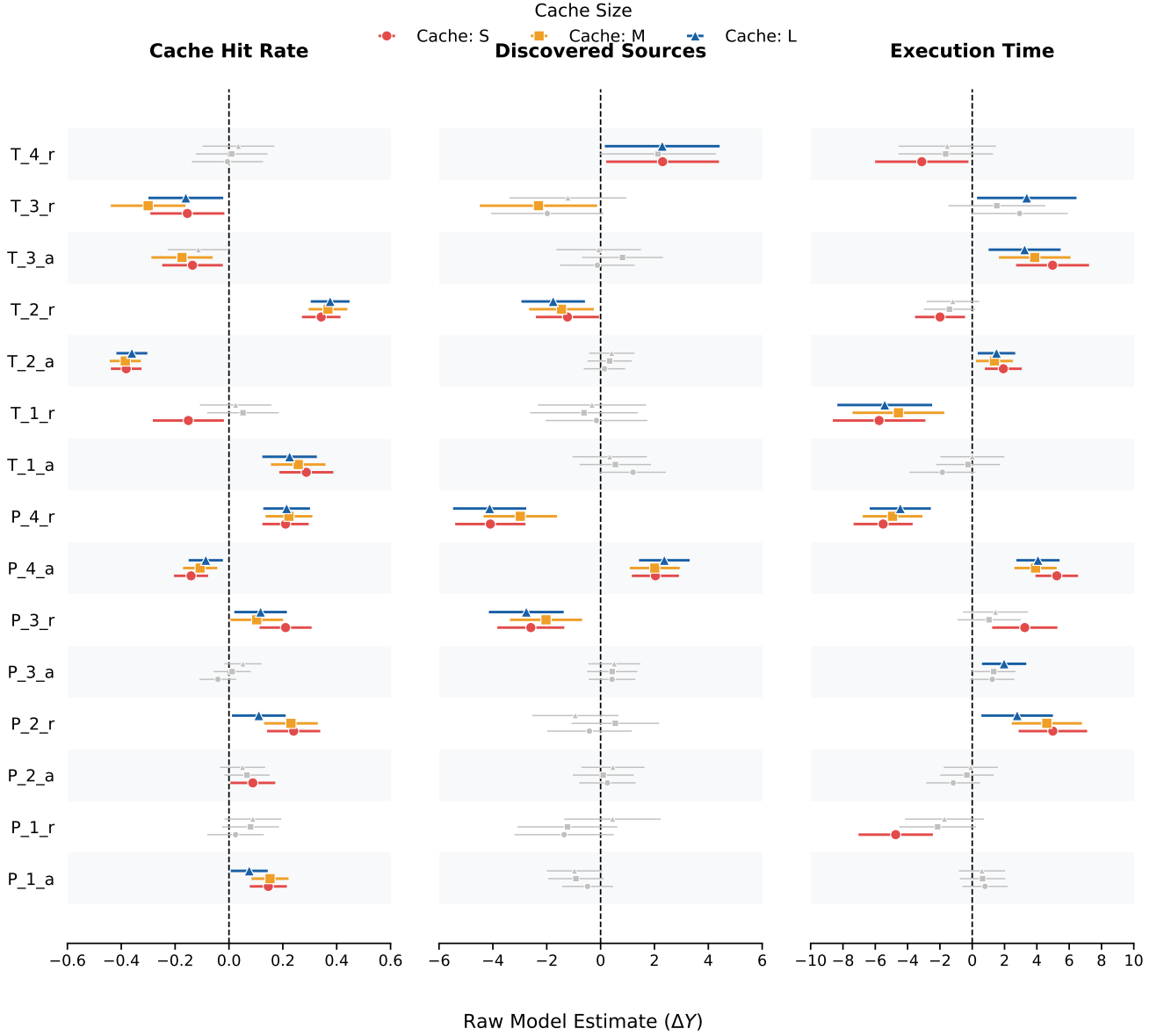


Figure 6: Marginal fixed effects of refinement pattern additions (-a) and removals (-r) on query performance across Small (S), Medium (M), and Large (L) cache capacities. The plot reports the unstandardized absolute expected change (ΔY) for Cache Hit Rate, Discovered Sources, and Execution Time. For both discovered sources and execution time, the plot uses signed log-counts and signed log-milliseconds, respectively. Error bars represent 95% confidence intervals and statistically insignificant estimates ($p \geq 0.05$) are rendered in gray. Undoing the traversal chokepoint changing the instantiation value of the query significantly improves cache hit rate (+.35) and reduces the number of unique data sources accessed. However, its effect on execution time is on the edge of significance.

significant negative coefficient of the lagged cache hitrate value in the regression model, which indicates that a higher previous hitrate reduces the expected change in hitrate.

Finally, we find that in general, chokepoints that significantly increase the number of sources accessed, significantly reduce cache hit rate and vice versa.

5.3.2. Effect of Refinement Depth on Performance

Next, using default query generation, assess the effect of refinement depth on cache performance, followed by implications for cache sizing and eviction policies.

5.4. Comparison to Alternative Evaluation Methods

The final step in our evaluation is to compare the performance outcomes obtained from our benchmark against alternative evaluation methods from literature. **{TODO: Mention random / synthetic}**

Compare SolidSessionBench's performance outcomes against alternative evaluation methods, specifically micro-benchmarks and random query sequences. Highlight differing performance characteristics, identify the analytical

shortcomings of these alternative methods, explain how their resulting conclusions mislead, and demonstrate how this sequence-based benchmark resolves those issues.

Microbenchmarks-Theoretical Hit Rate Analysis Cache literature extensively uses microbenchmarks of various forms to analyze caching performance [40, 26]. To compare the conclusions of such a benchmark to the conclusions of the benchmark introduced in this paper we investigate theoretical caching speedup without management costs, by executing each query template twice with simulated cache miss probabilities. We excluded the estimation-based approaches, as using the full cache for estimation would skew the results for miss rates above zero.

Figure 7 shows representative performance profiles for the workload. Most templates (8 out of 12) show reliable improvement that scales with the hit rate for both algorithms. However, certain long-running queries (e.g., interactive-discover-6, 7, and short-3) show no speedup, and some even experience slowdowns. These templates traverse many solid pods but do not require all traversed data. We hypothesize that rapidly serving documents from an in-memory cache overwhelms the system with data, slowing result production. Conversely, HTTP network latency gives the JavaScript event loop time to process the join execution pipeline, yielding initial results faster since only the first few requests are needed.

In general, using an indexed cache (store-based or LRU) improves performance more than an unindexed cache, however, this also comes with higher cache management costs, which are not accounted for when using theoretical hit rates. **{TODO: See about disk-base caching} {TODO: Circle back to our general evaluation to see how the conclusions differ}**

Random Query Sequences

Scope and Limitations We evaluate SolidSessionBench against microbenchmarks and random sequences, but explicitly exclude hand-crafted scenarios and simple distributions. Hand-crafted methods lack scalability, while simple distributions fail to capture complex user habits. By automating logical sessions and user interests, our benchmark functions as a scalable superset of both approaches, making direct comparison redundant. By contrast, the compared methods do not incorporate these sequence attributes, and our results show this omission risks drawing inaccurate conclusions.

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6. Conclusion

We introduced SolidSessionBench, an evidence-based benchmark for generating realistic query sequences in decentralized environments. By modeling logical sessions, user interests, and iterative refinement, it captures client-side performance characteristics that traditional evaluation methods fail to find. While using Solid as the primary use case, the underlying concepts apply to approaches such as Linked Web Storage (LWS), and we will provide an LWS-specific representation once the W3C Working Group finalizes the specifications.

To ensure longevity, SolidSessionBench has been integrated into the core SolidBench ecosystem, inheriting its established track record of community support and maintenance. We formalize this in our sustainability plan <https://github.com/SolidBench/SolidBench.js/wiki/Sustainability-Plan>. Due to the seamless integration into SolidBench, its status as the standard community resource for evaluating decentralized query execution [19, 61, 25, 59] serves as a proxy for the future adoption of this new benchmark when trying to evaluate any client-specific optimization approaches.

The framework is modular, MIT-licensed, and fully documented, with links to all code, data, and experiment setups available at this paper-specific hub: <https://github.com/RubenEschauzier/simulated-query-sequence-paper>. This extension-friendly approach to development allows researchers to easily modify or extend specific components to meet future requirements without disrupting the broader evaluation pipeline.

CRedit authorship contribution statement

Ruben Eschauzier: TODO. **Ruben Taelman:** TODO.

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Figure 7: Performance of unindexed, LRU indexed, In-memory store, and Disk-based store caches across representative query templates. The y-axis shows the mean speedup relative to a baseline with a 0% hit rate. Shaded regions denote the 90th percentile {TODO: Placeholder disk figure}

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